

Axy

Build guide for a handmade, whitish-pink wooden axolotl desk robot
with a Raspberry Pi brain, a talking AI and a mystical third eye

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Built by hand by Mat McDaniel · Design A (camera companion) with Design B fallback

October 2026 · All prices are estimates

1. Overview - meet Axy

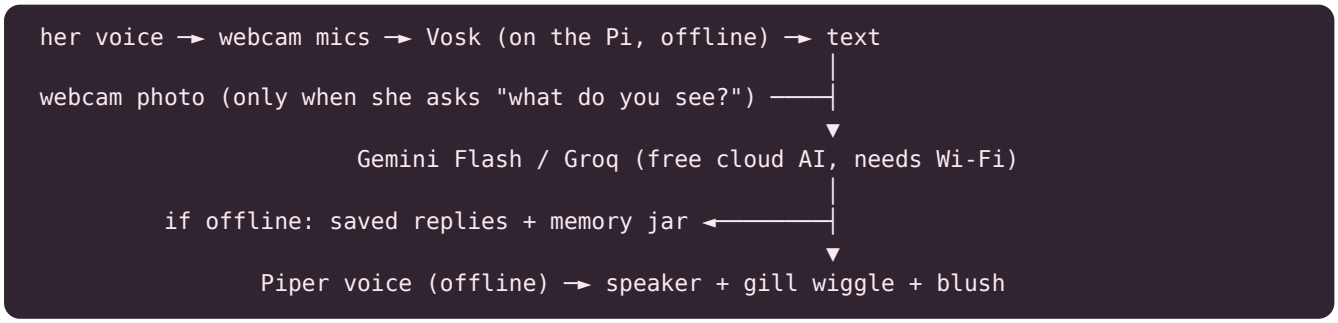
Axy is a chubby, whitish-pink wooden axolotl that lives on a desk. She has fluffy felt gills, two glowing eyes made from your LED headlamps, a mystical "third eye" in her forehead (your Logicool webcam), and a little speaker voice. You build her by hand; she is a gift for someone who loves axolotls.

This guide follows **LOCKED-PLAN.md**: one gift day and a finished robot. Nothing gets taken back for a "phase 2". **Design A** (camera companion) is the default. **Design B** (voice-only) is the automatic fallback if the webcam ever fails.

What she does on gift day

Feature	How it feels	How it works
Egg/nest reveal	She opens a painted egg and finds Axy sleeping in a nest. On first power-up Axy says *"Blub... I just hatched!"*	A one-time "hatch" greeting in the software
Push-to-talk chat	Hold the TALK button, speak, let go, and Axy answers out loud in 1-3 cute sentences	Vosk (offline speech-to-text) → free Gemini or Groq AI → Piper (offline voice)
"What do you see?"	Ask Axy what she sees and she peeks with her third eye: *"Ooh, you're holding a big mug and smiling!"*	One webcam photo goes to Gemini's free vision model
Looks at her	When she talks, Axy's head turns toward her face	The webcam photo + simple face detection + head servo
Gills wiggle + blush	Gills flutter and cheeks glow pink while Axy talks	2-3 SG90 micro servos + LEDs
Memory jar	Tap the MEMORY JAR button for a random voice clip recorded by Mat. A rare "jackpot" clip turns up now and then	Audio files on the SD card. Works with **no internet**
USB-C charge tail	Her phone charges from Axy's tail	Mat's power bank in the belly, with a USB-C cable out the tail
Secret belly note	A hidden note under the magnetic belly hatch	Paper and a magnet
Sleepy breathing eyes	The eyes slowly "breathe" when Axy is idle	PWM LEDs

How the pieces fit together



The big safety and reliability rules

1. **Servos get their own 5V supply.** Never power them from the Pi's 5V pin. Join the grounds. (Section 3 explains why.)
2. **Push-to-talk first.** It works far better than a wake word in a noisy room. A wake word can come later (Section 5.9).
3. **The memory jar never needs the internet**, so the gift still "works" even if her Wi-Fi is down.
4. **The API key is never written in the code.** It lives in a private file, `~/axy/axy.env`, read as an environment variable.

Build order (about 3 weekends, one gift)

1. **Weekend 1 - body:** cut, glue, sand and paint the body. Make the gills. Build the egg/nest. Record the memory-jar clips.
2. **Weekend 2 - electronics:** mount the Pi, wire the servos, eyes, buttons and speaker, fit the webcam third eye, run the power tail. Bench-test each part.
3. **Weekend 3 - brain and polish:** install the software, get the free API key, set the personality, do the full-system test and the gift-day dry run.

Hardware choices in one line each

- **Brain:** Raspberry Pi 4 (2-4GB) is best. A Pi 5 is fine (it has no headphone jack, so use a USB speaker). A Pi 3 works because the AI runs in the cloud.
- **Ears and third eye:** Mat's Logicool HD 1080p USB webcam with its built-in dual microphones.
- **Eyes:** Mat's two LED headlamps (their lenses and reflectors become shiny eyes, with a small LED inside each).
- **Motion:** 3 × SG90 micro servos: gills, head turn, tail wiggle.
- **Voice:** a small USB speaker (or a 3.5 mm one on a Pi 3/4).
- **Power:** the official Pi power supply for the Pi, a separate 5V for the servos, and Mat's power bank for the USB-C tail.
- **Network:** her home Wi-Fi, or Mat's Buffalo WMR-300 travel router as a ready-made backup.

2. Parts list and rough costs

> **All prices are rough estimates in yen (October 2026), not quotes from any shop.** Used prices depend a lot on luck (Mercari, Yahoo! Auctions, Hard Off, Jimoty, Sayonara sales). New prices vary by shop and by week. **This list is for information only. Nothing has been bought.**

A. Things Mat already owns (¥0)

Item	Used for
Logicool HD 1080p USB webcam (built-in dual mics)	Third eye (camera) **and** Axy's ears (microphone)
Power bank	Powers the USB-C charge tail for her phone
USB-C and micro-USB cables	The charge tail cable, plus a spare to cut into a servo power lead
LAN cables	Wired internet backup (Pi ↔ router or WMR-300)
3.5 mm / RCA audio adapters	Connecting a 3.5 mm speaker on a Pi 3/4
Buffalo WMR-300 travel router	Backup Wi-Fi so Axy can be set up before visiting her place
Two LED headlamps	Axy's eyes (lenses and reflectors, plus an LED inside each)

B. The brain (choose one)

Item	Qty	Used (¥)	New (¥)	Notes
Raspberry Pi 4 Model B, 2GB or 4GB (preferred)	1	5,000-12,000	9,000-16,000	Sweet spot. Has a 3.5 mm jack. 4GB+ also allows the optional offline AI
or Raspberry Pi 5, 4GB+	1	10,000-16,000	13,000-22,000	Fine and fast. **No 3.5 mm jack** (use a USB speaker). Needs a 5A supply for full USB power
or Raspberry Pi 3B / 3B+	1	3,000-6,000	6,000-10,000	Works, because the AI is in the cloud. A bit slower to speak. No offline AI
microSD card, 32GB, A1/A2 class	1	buy new	600-1,500	Don't use an old card. They corrupt
Official Raspberry Pi USB-C power supply	1	500-1,500	1,200-3,000	Pi 4: 5.1V 3A. Pi 5: 5.1V 5A (27W). Pi 3: micro-USB 5.1V 2.5A
Heatsinks / small case / Pi 5 active cooler	1	0-500	300-1,500	Axy lives in a closed wooden body, so cooling matters

Look for boards with the 技適 (Japanese radio certification) mark, so Wi-Fi use in Japan is legal.

C. Motion, lights and wiring

Item	Qty	Used (¥)	New (¥)	Notes
SG90 micro servo (comes with horns and screws)	3	0-1,500 total	1,200-3,600 total	Gills, head turn, tail. Buy one spare if cheap

Item	Qty	Used (¥)	New (¥)	Notes
5V servo supply: a spare USB phone charger, 5V 2A+	1	0 if owned	500-1,200	**Separate** from the Pi's supply
USB-to-bare-wire lead (or cut an old USB cable)	1	0 (cut a spare)	200-800	Red = +5V, black = GND. Alternative: a 4×AA box with NiMH cells (¥110-330 + cells)
Electrolytic capacitor 470-1000 µF, 10V or higher	1	-	30-150	Smooths servo power spikes
5 mm LEDs, pink or warm orange (small pack)	4+	-	100-400	2 eyes + 2 blush cheeks
Resistors 220 Ω (and a few 100 Ω)	6+	-	50-300	One per LED
Push buttons (12 mm tactile, or a cute 24 mm round button)	2	-	50-600 each	TALK and MEMORY JAR
Jumper wires (female-female + male-female, 40 pack)	1	0-300	200-700	No soldering needed for a first build
Mini breadboard or small perfboard	1	-	150-600	A "servo power strip" inside the belly
Heat-shrink tube or electrical tape	1	-	110-300	Insulate every joint
Optional: 2SC1815 / 2N2222 NPN transistors + 1 kΩ resistors	2	-	20-100	Only if you keep the headlamps' own bright LEDs (Section 3.4)
Optional: PCA9685 16-channel servo board	1	-	500-1,500	Smoother servos later. Not needed for this guide

D. Sound

Item	Qty	Used (¥)	New (¥)	Notes
Small speaker: USB-powered mini speaker (USB audio **or** 3.5 mm input)	1	0-800	550-2,500	Small enough to fit inside the body (~5 cm). USB-audio type works on every Pi
USB sound adapter (USB → 3.5 mm)	0-1	0-500	300-1,500	Only needed for a 3.5 mm speaker on a **Pi 5**
Backup USB microphone	0-1	0-800	500-1,800	Optional. Only if the webcam mic turns out muffled

E. Body and craft (Daiso / Seria - mostly ¥110 items)

Item	Qty	Estimate (¥)	Notes
Unfinished wooden box, about 15 × 10 × 8 cm (torso)	1	110-550	Must hold the Pi + power bank. Measure your bank first!
Round wooden box, ball or thin boards (head, about 10-12 cm wide)	1-2	110-330	Must fit the webcam behind the forehead
Thin craft board / balsa (tail, legs, gill bar)	1-2	110-220	
Felt sheets: white, pale pink, coral	3-4	330-440	Gills, cheeks, spare gills
Pipe cleaners (ﾓｰﾙ), pink/white	1 pack	110	Wire cores for the gill fronds

Item	Qty	Estimate (¥)	Notes
Acrylic paint: white, pink, a touch of coral/red, black	3-4	330-550	Mix "sakura milk" whitish-pink
Matte clear varnish / top coat	1	110-220	Stops paint chipping when she holds Axy
Wood glue, sandpaper (#120 + #240)	1 each	220	
Hot-glue gun + sticks	1	220-550	
Small strong magnets (belly hatch)	1 pack	110	
Felt or rubber furniture feet	1 pack	110	Protects her desk
Mini lazy-susan turntable **or** 2 washers + bolt	1	110-550	Carries the head's weight so the servo only turns it
1/4-inch camera screw (1/4"-20) or Velcro	1	110-400	Fixes the webcam by its tripod thread
Cable ties, double-sided foam tape	1 each	220	
Craft subtotal		**≈ 2,500-5,000**	

F. Egg / nest reveal

Item	Estimate (¥)	Notes
Papier-mâché egg / round box, or a wooden box painted like an egg	110-550	Big enough for Axy (~20 cm)
Shredded paper / felt moss / raffia "nest"	110-220	
Battery fairy lights (optional)	110-330	
Tiny glass jar + paper for rolled "memories" (optional)	110	The physical memory jar, for the ceremony
Card for the inside-lid note and the belly note	110	
Egg subtotal	**≈ 440-1,300**	

G. Tools (borrow if you can)

Small Phillips screwdriver, craft knife, small hand saw, hand drill or awl (2-10 mm holes), ruler, pencil, masking tape, paintbrushes, and optionally a soldering iron (≈ ¥1,500-4,000 new). Everything in this guide can be done **without soldering**, using jumper wires, heat-shrink and tape. Soldering just makes it sturdier.

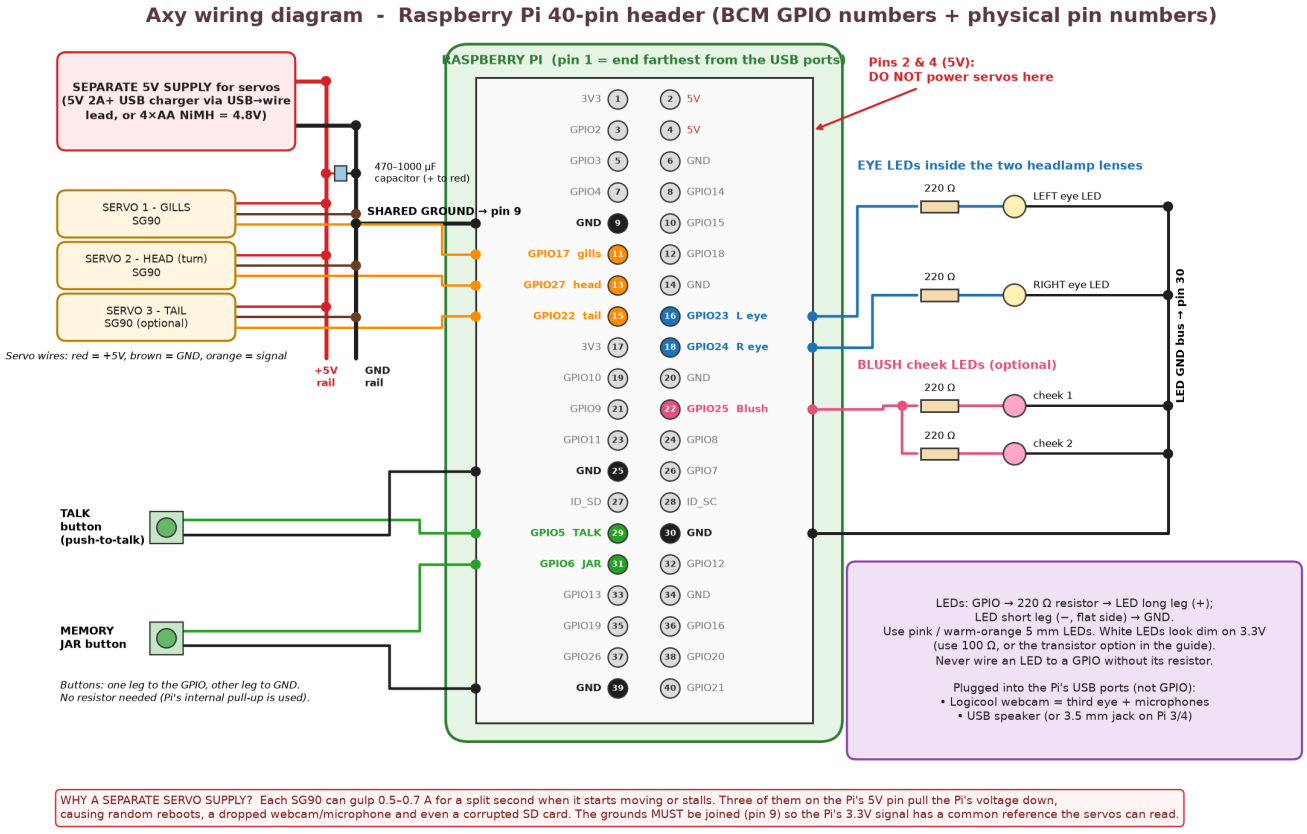
Rough total (estimates only)

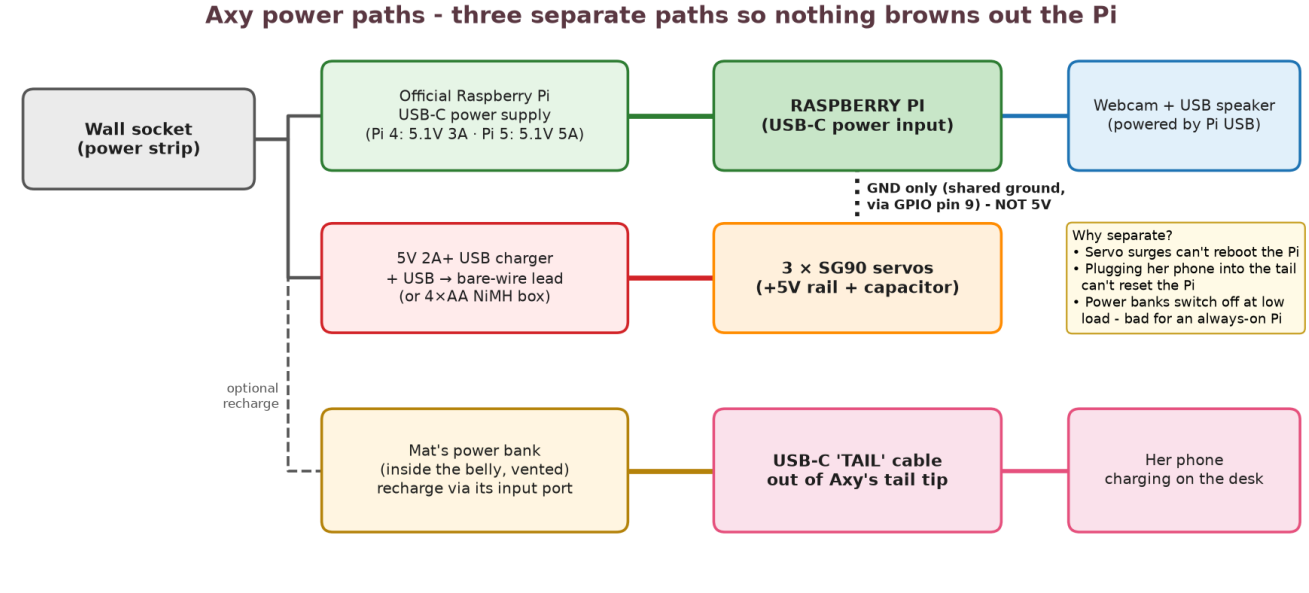
Route	Estimate
Used-hunting route (used Pi 4, scavenged speaker and charger)	**≈ ¥10,000-25,000**

Route	Estimate
All-new route with a Pi 4	**≈ ¥17,000-35,000**
All-new with a Pi 5	add roughly ¥4,000-8,000

The webcam, power bank, cables, router and headlamps you already own save a good chunk of this.

3. Wiring diagram and pin table





3.1 Pin table

Two kinds of numbers: **GPIO numbers** (what the software uses, e.g. GPIO17) and **physical pin numbers** (count along the header, 1-40). Pin 1 is at the end of the header **farthest from the USB ports**. Odd pins (1, 3, 5…) are the row toward the middle of the board. Even pins (2, 4, 6…) are the row along the board's edge. To double-check, run `pinout` on the Pi: it prints a map of your exact board.

Part	Wire	GPIO (software)	Physical pin	Other end goes to
Servo 1 - **gills**	orange (signal)	**GPIO17**	**11**	-
Servo 2 - **head turn**	orange (signal)	**GPIO27**	**13**	-
Servo 3 - **tail wiggle**	orange (signal)	**GPIO22**	**15**	-
All 3 servos	red (+5V)	-	**NOT the Pi**	**+5V of the separate servo supply**
All 3 servos	brown (GND)	-	-	GND rail of the servo supply
Shared ground	black	GND	**9**	GND rail of the servo supply
TALK button (push-to-talk)	leg 1	**GPIO5**	**29**	leg 2 → GND **pin 25**
MEMORY JAR button	leg 1	**GPIO6**	**31**	leg 2 → GND **pin 39**
Left eye LED	via 220 Ω → LED long leg	**GPIO23**	**16**	LED short leg → LED GND bus → **pin 30**
Right eye LED	via 220 Ω → LED long leg	**GPIO24**	**18**	LED short leg → LED GND bus → **pin 30**
Blush cheek LEDs (×2, optional)	each via its own 220 Ω	**GPIO25**	**22**	short legs → LED GND bus → **pin 30**
Webcam (camera + mics)	USB	-	any USB port (blue USB 3 on a Pi 4/5)	-
Speaker	USB, or 3.5 mm jack (Pi 3/4)	-	USB port / AV jack	-
Pi power	USB-C (Pi 4/5) / micro-USB (Pi 3)	-	power input	official Pi power supply

These pin numbers are already set in `software/config.toml` under `[gpio]`. If you use different pins, change them there. The config uses GPIO numbers, **not** physical pin numbers.

Pins to leave alone: pins 2 and 4 (5V) and 1 and 17 (3.3V) are **not used at all** in this build.

3.2 Why the servos need their own 5V supply (important)

An SG90 is tiny, but every time it starts moving, or pushes against something (a stall), it gulps a burst of current: roughly 0.5-0.7 A for a split second. Three servos wiggling at once can ask for well over 1.5 A in spikes.

If those spikes come from the Pi's 5V pin, they come out of the same supply that runs the Pi's processor, Wi-Fi and USB ports. The voltage dips, and you get:

- **random reboots or freezes**, usually right when Axy starts talking (because that's when the gills move);
- **the webcam or speaker dropping off USB** for a moment, which sounds like a crackle or a "lost microphone";
- in the worst case, a **corrupted SD card** if the power dips while the Pi is writing.

So the servos' **red wires go to a separate 5V supply** (a spare 5V 2A+ USB phone charger with a USB-to-bare-wire lead, or a 4×AA NiMH battery box = 4.8V). The Pi has its own official supply.

Join the grounds. The servo supply's GND (black/brown) must also connect to a Pi GND pin (pin 9). The servo reads the Pi's 3.3V signal *relative to ground*. Without a shared ground, the signal has no reference and the servos twitch randomly or don't move at all. You join **only GND**, never the two +5V lines.

Add a **470-1000 µF capacitor** across the servo supply's +5V and GND, close to the servos. The **stripe on the capacitor marks the minus (−) leg**, which goes to GND. It acts like a tiny water tank for the current spikes.

3.3 Making the servo power strip (no soldering)

1. Take an old USB-A cable you don't need (or a ready-made "USB to bare wire" lead). Cut off the far end and strip back the outer jacket about 3 cm.
2. Inside you'll usually find red (+5V), black (GND), and white/green (data). **Cut the white and green short and tape them off.** Strip 5 mm off the red and black.
3. Check with a multimeter if you have one: plug into the charger, and red to black should read about 5V.
4. Push red and black into two rows of a mini breadboard (mark them **+5V** and **GND** with a pen). Insert the capacitor across those rows (stripe/minus to GND).
5. Plug each servo's 3-wire plug into male-female jumpers: **red → +5V row**, **brown → GND row**, **orange → long jumper to the Pi pin** in the table.
6. One extra jumper runs from the **GND row → Pi pin 9**. That's the shared ground.

Can't find a spare charger? A **4×AA battery box with rechargeable NiMH cells (4.8V)** works great and is very "clean" power. Don't use 4 alkaline cells: 4 × 1.5V = 6V is above the SG90's comfortable range.

3.4 The eyes: using your LED headlamps

The headlamps give Axy beautiful eyes: a round **reflector + clear lens**, like a glossy glass eye. There are two ways to light them:

Option A - recommended, easy and safe. Take the headlamp front apart (usually a few screws or clips). Keep the reflector, lens and bezel. Put a **5 mm pink or warm-orange LED** where the headlamp's LED was, with a **220 Ω resistor** on its long leg, and wire it as in the diagram (GPIO23/GPIO24). The software can then "breathe", blink and glow the eyes. - **LED legs:** the long leg is **+** (to the resistor and then the GPIO). The short leg, or the flat side of the rim, is **−** (to GND). - Paint a black "pupil" dot in the middle of the lens, or glue a black felt circle with a clear center, and the glow becomes a cute ring. - White LEDs look dim from a 3.3V GPIO. If you want white, use 100 Ω, or use Option B.

Option B - keep the headlamp's own LED (for experienced builders). Headlamp LEDs are often 1 W "power" LEDs. They can draw far more than a GPIO pin can give (about 16 mA max) and **must not be wired straight to a GPIO**. If you want to keep it, switch it with an NPN transistor (2SC1815 or 2N2222): - GPIO → **1 kΩ** → transistor **base**; transistor **emitter** → GND; transistor **collector** → LED **−**; LED **+** → a current-limiting resistor (about 100 Ω for a small 3V LED, from 5V) → **servo +5V rail**. - 2SC1815 legs (flat face toward you, legs down): **E C B** from left to right. Check your transistor's datasheet, because legs differ between parts. - If you keep the headlamp's battery pack instead, its minus must join the shared GND. Honestly, Option A is simpler and looks just as good.

3.5 Buttons

Each button has two legs that connect when pressed. One leg goes to the GPIO pin, the other to a GND pin. No resistor is needed: the software turns on the Pi's built-in "pull-up". Tactile buttons often have **4 legs** in two pairs that are already joined inside. Use two legs that are **diagonal** from each other and you can't get it wrong.

- **TALK** (GPIO5, pin 29): put this one where she naturally presses, such as the top of the head or the snout.
- **MEMORY JAR** (GPIO6, pin 31): put this on the back or the side, maybe under a felt "heart" patch.
- **Tap** = random memory clip.
- **Hold 2 seconds** = Axy speaks her status (online/offline, IP address, camera, mic, key). This is very handy for Mat.

3.6 Before you power on (every time you change wiring)

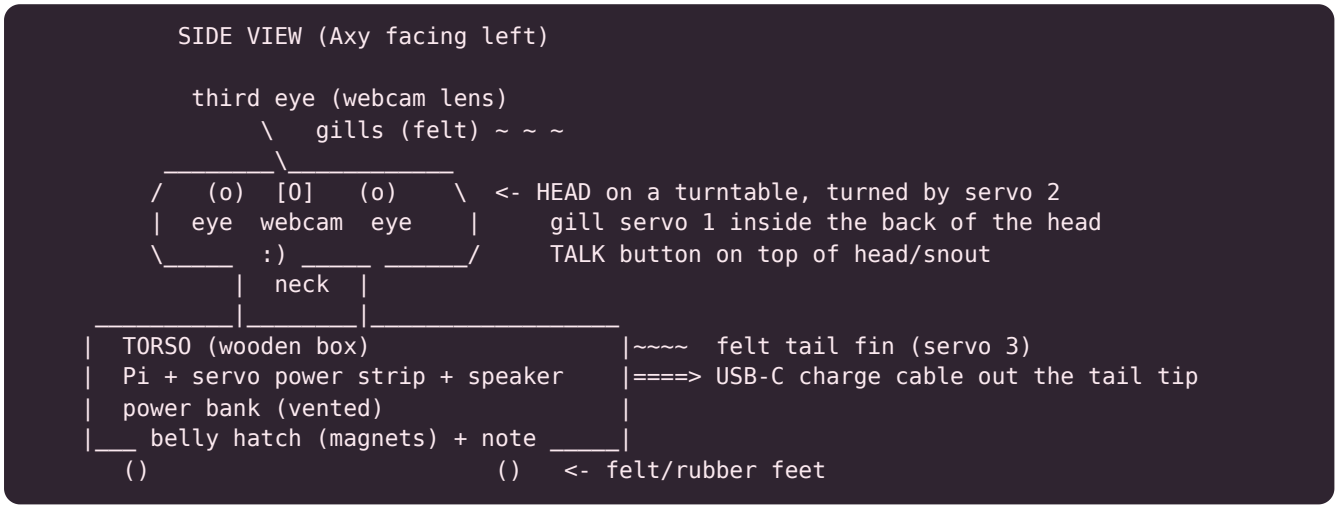
- [ ] No servo red wire touches the Pi's pins 2 or 4.
- [ ] The servo supply GND is joined to Pi pin 9.
- [ ] Every LED has its own resistor.
- [ ] No bare metal touches other bare metal: tape or heat-shrink every joint.
- [ ] The capacitor's stripe (−) goes to GND.
- [ ] Power the **Pi first**, then the **servo supply**. When turning off, shut down the Pi properly (`sudo shutdown now`) before pulling its plug.

4. Step-by-step assembly

Take your time. Paint needs a night to dry, and a slow, tidy build is what makes the gift feel special. **Photograph each stage.** If something goes wrong later, the photos show you where every wire went.

4.0 Layout and sizes

Target size: **about 18-22 cm long, 10-13 cm tall** with the head on. Measure your **power bank** and **webcam** before you buy the boxes, because they set the minimum size.



4.1 Body (torso)

1. **Choose the torso box**, about 15 × 10 × 8 cm. The open side becomes the **belly hatch** (the bottom). Check that the Pi (8.5 × 5.6 cm), the power bank and the speaker all fit inside with room for wires.
2. **Round it off.** Sand the corners and edges into soft curves (#120 sandpaper, then #240). Axolotls are chubby. Optionally glue thin balsa or layered card on the sides and sand it into a plump "cheek" shape.
3. **Mark and cut the holes** before painting: - **Neck hole** in the top, near the front: 3-4 cm across, for the head servo and cables. - **Tail hole** in the back: about 1.5 cm, for the USB-C charge cable and tail servo wires. - **Rear cable hole**: 1.5 cm, low on the back, for the Pi power cable and the servo supply lead. - **Speaker grille**: 15-20 small holes (3 mm) in a circle or heart shape on the chest or side. - **Vent holes**: a row of 5 mm holes on the belly hatch and low on the back. Warm air must escape. - **Memory-jar button hole** on the back or side (button diameter).
4. **Legs:** four stubby legs from 1.5 cm dowel or rounded wood blocks, glued under the corners. Make them just tall enough (about 1 cm) to lift the belly vents off the desk.
5. **Belly hatch:** cut a board to fit the open bottom. Glue small magnets in the 4 corners of the box and matching magnets (check polarity!) on the hatch. Glue felt feet under the legs.
6. **Paint:** mix **white + a small dot of pink** into a soft "sakura milk" whitish-pink. Do 2-3 thin coats, drying between coats. Add blush-pink cheeks and a few tiny darker spots. Paint inside edges too. Finish with **matte varnish**. Let it dry overnight before fitting electronics.
7. **Secret belly note:** glue a small paper pocket inside the hatch. The note goes in on gift day. Optionally write the gift date and "made by Mat" under the belly.

4.2 Gills (the star of the show)

Axolotls have **three feathery gill stalks on each side** at the back of the head.

1. Cut **6 frond shapes** from pale-pink felt, about 4-6 cm long, like a long feather. Cut 6 more as backs.
2. Bend a **pipe cleaner** into each frond shape, leaving a 2 cm "stem" sticking out.
3. Sandwich each pipe cleaner between two felt pieces with hot glue (or tacky glue). Snip a **fringe** along both edges, and brush coral paint or glue coral felt on the **tips**.
4. Make the **gill bar**: a craft stick (popsicle stick) about 10-12 cm long, painted to match. Glue **3 fronds to each end**, fanned out top / middle / bottom.
5. **Gill servo (servo 1):** fix it inside the **back of the head** with hot glue or a small wooden bracket, with the shaft pointing **backwards** through a 1 cm hole. Before fitting the horn, run `axy.py --test servos` (Section 7) so the servo is at its **centre** (0°). Then screw a straight horn on, horizontal, and glue the gill bar to the horn.
6. Bend the pipe cleaner stems so the fronds curve forward around the sides of the head. When the servo rocks ±20°, one side lifts while the other dips: a gentle flutter.
7. Make a **spare set**. They get loved to death.

4.3 Eyes (LED headlamps)

1. Take the two headlamps apart. Keep each **bezel + reflector + clear lens** as one "eye unit". Put the batteries and straps aside.
2. Mark two eye holes on the front of the head, **wide apart and slightly to the sides** (axolotl style), a little above the smile. Make the holes just the size of the reflector so the bezel sits on the surface. Drill a small hole and enlarge it with a craft knife or a round file.
3. Fit a **5 mm pink or warm-orange LED** in the centre of each reflector (where the old LED was) with hot glue. Solder or crimp a **220 Ω resistor** to the **long leg** and slip heat-shrink over it. Attach a female jumper to each end: resistor end → GPIO, short leg → GND. (See Section 3.4 if you want to keep the headlamp's own LED.)
4. Optional: paint or glue a dark "pupil" ring on the inside of the lens so the eye looks cute, not like a torch. Test the brightness (Section 7) **before** gluing the eye in.
5. Press each eye unit into its hole and hot-glue it from behind. Label the wires **L EYE** and **R EYE** with masking tape.
6. **Blush cheeks (optional):** a 5 mm pink LED behind a thin spot of felt or a 5 mm hole on each cheek, each with its own 220 Ω resistor, both on GPIO25.

4.4 Third-eye webcam mount (forehead)

The Logicool webcam is quite wide (roughly 9-10 cm), so build the head around it.

1. **Test the webcam first** on a computer or on the Pi (`axy.py --test camera` and `--test mic`, Section 7) so you know the picture and the mic both work.
2. **Head shell:** use a round wooden box or a hollow shape about **11-13 cm wide**. The webcam sits inside, behind the forehead, centred between and **above** the eyes.
3. **Lens hole:** drill a hole in the centre of the forehead **about 2.5-3 cm across**, exactly in front of the lens. Check from behind that no edge of the hole shows in the picture (take a test photo).
4. **Microphone holes:** the webcam's two mics are small openings on its **front face, either side of the lens** (look for pinholes or a fabric strip). Drill **2-3 small holes (3 mm)** in front of each one. A muffled mic is the most common webcam problem, and wood over the mics makes it worse.
5. **Fixing it:** most Logicool webcams have a **1/4-inch tripod thread** in the clip. Screw it to a small wooden shelf inside the head with a 1/4"-20 camera screw. Otherwise fold the clip flat and use

Velcro plus hot glue on the **clip only** (never on the lens or mic area). The webcam must not wobble.
6. **Make it mystical:** around the lens hole, paint a soft lavender or pink ring, add tiny dots or a little gold "third eye" lash pattern, or glue a felt ring. Keep everything **out of the lens's view**.
7. Route the webcam's USB cable down through the neck hole with a **loose loop** of slack, so the head can turn ±35° without tugging. Tie the loop gently with a cable tie inside the torso.
8. **Lighting tip for later:** the camera works best when she isn't sitting in front of a bright window.

4.5 Servos (head turn and tail)

Before attaching any horn, centre every servo by running `axy.py --test servos` once. Each servo ends at 0°. Then fit the horns pointing "straight".

Head turn (servo 2, GPIO27): 1. On the torso top, around the neck hole, fit a **mini lazy-susan turntable**, *or* glue **3 slippery furniture glide pads** in a triangle. These carry the head's weight, so the little servo only has to turn it, not hold it up. 2. Mount the servo **under the torso lid**, shaft pointing **up** through the centre of the neck hole. Use a small wooden bracket and hot glue or 2 small screws. 3. Glue/screw a round or cross horn to a **head base plate** (a wooden disc about 7-8 cm). Press the horn onto the servo shaft and screw it on. The plate rests on the turntable/glides. 4. Glue the head onto the base plate, facing straight ahead when the servo is centred. 5. Feed all head wires (webcam USB, gill servo, eye LEDs, talk button) down through the neck with slack. Turn the head by hand gently, ±35°, and check nothing pulls. 6. If the head turns the wrong way when following her face, set `head_invert = true` in `config.toml`.

Tail (servo 3, GPIO22): 1. Make a **fixed tail root**: a short hollow tube (wood, card or foam) glued to the back over the tail hole. The **USB-C charge cable runs through the root and out of the tip**, so the cable itself never wiggles. 2. Make a **soft tail fin** from felt over thin balsa (about 8 cm, a long leaf shape) and attach it to the tail servo's horn. Mount the servo in the rear of the torso, with its shaft poking out above the root, so the fin waves gently (±15°) along the top of the tail. 3. Don't want a tail servo? Set `servo_tail = -1` in `config.toml`. Some builders use this 3rd servo for a **head tilt/nod** instead (LOCKED-PLAN.md listed pan + tilt + gills). Same wiring, just mount it differently.

4.6 Speaker

1. Choose the speaker position behind the **speaker grille holes** you drilled (chest, or belly side if the legs lift it).
2. Glue a scrap of thin fabric or felt behind the holes (it stops dust and hides the speaker), then fix the speaker with hot glue or double-sided foam tape, facing the holes.
3. **USB speaker:** plug it into a Pi USB port. **3.5 mm speaker:** plug the audio into the Pi 3/4 headphone jack (your 3.5 mm / RCA adapters help here) and its power into a USB port. **Pi 5:** use a USB speaker or a USB sound adapter.
4. Don't press the speaker right against the webcam: the head and torso are separate, which is good.
5. Set a gentle default volume (Section 5.6). Soft is cuter, and it keeps the speaker from rattling.

4.7 Buttons

1. **TALK button** on top of the head or on the snout: drill a hole, glue the button from inside so only the cap pokes out, and paint or felt-cover the cap (a little pink "bubble"). Run its 2 wires down the neck.

2. **MEMORY JAR button** on the back or side of the torso. Cover it with a felt heart, or a tiny painted "jar" symbol.
3. Connect as in the pin table: TALK → GPIO5 (pin 29) and GND (pin 25). JAR → GPIO6 (pin 31) and GND (pin 39).

4.8 Power and the USB-C charge tail

1. Lay the **power bank** flat at the back of the torso, on a strip of felt, with its ports reachable from the belly hatch. **Don't wrap it tightly.** Keep space and vents around it.
2. Plug a **USB-C cable** (from your own cables) into the bank's output, run it through the **tail root**, and leave 25-40 cm out of the **tail tip** for her phone. Hot-glue a small loop of the cable inside the body as **strain relief**, so a pull on the tail can't yank the plug.
3. **Check her phone's connector!** Recent iPhones and most Android phones use USB-C, but an older iPhone uses Lightning. Then you'd want a USB-C → Lightning cable (or an adapter) in the tail.
4. Recharging the bank: either open the belly hatch now and then, or run a short USB extension from the bank's **input** port to the rear cable hole. If your bank supports **pass-through** (charging while charging), it can stay plugged in. If not, don't leave it plugged in permanently.
5. **Pi power:** the official Pi supply's cable enters through the **rear cable hole**, separate from the tail.
6. **Servo power:** the 5V USB charger lead also enters through the rear hole, to the servo power strip. Bundle the two wall cables in a pink cable sleeve or wrap them in felt "seaweed".
7. Why not run the Pi from the power bank? Many banks briefly cut their output when a phone is plugged in, or switch off at low load. That would reboot Axy mid-sentence. The bank is only for the tail. (For a quick **battery-only demo**, a bank rated 5V 3A can run a Pi 4 for a while, but expect occasional resets.)

4.9 Fitting the electronics bay

1. Mount the **Pi** on a scrap board with nylon standoffs, or with hot glue on its **edges only**. Then fix the board in the torso with the USB ports reachable and the SD card slot accessible.
2. Fix the **servo power strip** (mini breadboard) beside it with foam tape.
3. Make every connection in the pin table, label each wire with masking tape, and tidy with cable ties.
4. Keep the Pi's heatsink area open to air, and don't let the power bank sit directly on the Pi.
5. Run through the **"Before you power on" checklist** in Section 3.6, then the bench tests in Section 7.
6. Close the belly hatch (magnets) only after everything passes.

4.10 Egg / nest reveal

1. **Egg:** a papier-mâché egg, a large round box, or a wooden box, painted in the same whitish-pink with soft darker spots. It should be big enough for Axy plus nest material.
2. **Nest:** shredded paper or felt moss and raffia. Optional battery fairy lights tucked inside for a glow when she opens it.
3. **Inside the lid:** a handwritten note and a tiny photo of you two.
4. **Physical memory jar (optional):** a little glass jar in the nest with rolled paper "memories". The real jar is the button, so this one is just ceremony.
5. **Reveal sequence:** Axy is in the nest, the power cable is hidden under the nest material, and the egg is closed. She opens the egg, Axy "hatches" (the first-boot greeting), and Mat says *"press the*

heart on her back" (memory jar first), then *"now hold the button on her head and say hi."*

6. Plan the cable: either Axy is already plugged in under the nest (her eyes breathing softly when the lid opens is magical), or you plug her in as the "hatching" moment. She takes about 40-60 seconds to boot.

5. Software setup

Everything here is free. You'll type commands into the Pi over **SSH** (a remote terminal) from your laptop, so the Pi doesn't need a monitor or keyboard.

Files in `software/`:

File	What it is
`setup.sh`	One-shot installer: packages, Vosk, Piper, voice models, start-on-boot service
`axy.py`	Axy's program: buttons, servos, eyes, listening, AI, voice, memory jar, fallbacks, self-tests
`config.toml`	All the settings: her name, personality, pins, models, volumes
`axy.env.example`	Template for the private file that holds your free API key
`axy.service`	systemd unit so Axy starts on boot (installed by `setup.sh`)
`memory-jar/README.txt`	How to name the memory clips

5.1 Flash Raspberry Pi OS

1. On your computer, install **Raspberry Pi Imager** from raspberrypi.com/software.
2. Insert the microSD card. In Imager choose: - **Device:** your Pi model (4, 5 or 3). - **OS:** *Raspberry Pi OS (other)* → **Raspberry Pi OS Lite (64-bit)**. No desktop: it's lighter, boots faster, and keeps the audio simple. Choose 64-bit even on a Pi 3, because the Piper voice needs it. - **Storage:** your microSD card.
3. When it asks about **OS customisation**, choose *Edit settings*: - **Hostname:** `axy` - **Username / password:** e.g. `mat` and a password you'll remember. - **Wi-Fi:** your home Wi-Fi name and password, **Wireless LAN country: JP**. - **Locale:** time zone `Asia/Tokyo`, keyboard `jp` or `us` (whichever you use). - **Services:** tick **Enable SSH** (use password authentication).
4. Write the card, put it in the Pi, and power on. The first boot takes a few minutes.
5. From your laptop's terminal (Windows: PowerShell. Mac/Linux: Terminal): `bash ssh mat@axy.local` Type `yes` the first time, then your password. If `axy.local` isn't found, look up the Pi's IP address in your router's device list and use `ssh mat@192.168.x.x`.
6. Update the system once: `bash sudo apt update && sudo apt full-upgrade -y && sudo reboot`

5.2 Copy the software and run the installer

From your **laptop**, in the `build-docs` folder:

```
scp -r software mat@axy.local:~/axy-software
```

Then on the **Pi** (over SSH):

```
cd ~/axy-software
bash setup.sh
```

The installer takes 10-20 minutes. It: 1. installs system packages (audio tools, ffmpeg, gpiozero, OpenCV for face-follow, espeak-ng as an emergency voice); 2. copies `axy.py` and `config.toml` into

`~/axy/`, and creates a private `~/axy/axy.env` (only you can read it); 3. makes a Python environment in `~/axy/venv` and installs **Vosk** (speech-to-text) and **Piper** (voice); 4. downloads the small English Vosk model (~40 MB) and the Piper voice **en_US-amy-medium** (~60 MB); 5. adds you to the `gpio`, `audio` and `video` groups; 6. installs and enables the `axy` **systemd service**, so Axy starts on every boot.

Options: `bash setup.sh --no-service` skips the boot service. `bash setup.sh --with-ollama` also installs the optional offline brain (Section 6.3, Pi 4/5 with 4GB+ only).

Log out and back in (or `sudo reboot`) afterwards so the new group permissions apply. Running `setup.sh` again later is safe: it keeps your config, key and clips.

> **The LLM client** is built into `axy.py`. It talks to Gemini and Groq with plain HTTPS using Python's standard library, so there's no extra SDK to install or break. The model names live in `config.toml` (`[brain]`), so when Google or Groq rename models you only edit one line.

5.3 Get a free API key (pick one, or get both for backup)

Option 1 - Google Gemini (recommended: chat and "what do you see?")

1. Go to **https://aistudio.google.com/apikey** and sign in with a Google account.
2. Accept the terms, then click **Create API key** (it may ask you to create or choose a project; the default is fine).
3. Copy the key. **Don't add billing.** Axy is designed for the free tier.
4. Defaults in `config.toml` (October 2026): chat `gemini-3.5-flash-lite` (fast), vision `gemini-3.8-flash`. If Google renames models, see the list at ai.google.dev/gemini-api/docs/models and change those two lines.

Option 2 - Groq (very fast chat)

1. Go to **https://console.groq.com/keys**, sign up and log in.
2. **Create API Key**, give it a name like `axy`, and copy it now (it's shown only once).
3. Defaults: chat `openai/gpt-oss-20b`, vision `qwen/qwen3.8-27b`. Model list: console.groq.com/docs/models.

Put the key in Axy (never in the code)

```
nano ~/axy/axy.env
```

Replace the placeholder on the matching line, with no quotes and no spaces:

```
GEMINI_API_KEY=AIza...your-key...
```

Save with **Ctrl+O, Enter**, exit with **Ctrl+X**. Then pick the provider in `~/axy/config.toml` (`provider = "gemini"` or `"groq"`). If you have both keys, Axy automatically tries the other provider when the first one fails.

Keep the key secret. Don't post it, put it in a photo, or commit it to GitHub. If it leaks, delete it in AI Studio / the Groq console and make a new one.

Free-tier notes: limits (requests per minute and per day) change over time and are shown in AI Studio / the Groq console. A desk pet uses very little, but if Axy says *"My brain bubbles are a bit busy"*, you've briefly hit a limit. Google's free tier may use what you send to improve its products, so Axy's prompt avoids private details, and only short sentences and (when asked) one photo are sent.

5.4 Set Axy's personality

Open the config:

```
nano ~/axy/config.toml
```

1. Under `[people]`: set `her_name`, and replace the `sweet_facts` with 2-4 short, kind, private touches (favourite snack, a nickname, a shared place). Axy may mention them gently now and then.
2. Under `[personality]`, the `system_prompt` already describes a **cute, gentle axolotl friend**: soft, affectionate, a little sleepy and silly, 1-3 short sentences, uses her name sometimes, never creepy, kind about anything it sees. Ideas to tweak it: - Add a catchphrase: *"Sometimes end with a tiny 'blub'."* - Add your inside joke: *"If she says 'potato', remind her of the Kyoto noodle day."* - Make it bilingual: *"If she speaks Japanese, answer in simple Japanese."* (The voice and ears are English by default. See "unsure" notes in Section 8.)
3. `[offline_replies]` holds a few keyword → reply pairs Axy uses when there's no internet. Add your own.
4. Restart to apply: `sudo systemctl restart axy`.

Axy remembers the last 6 back-and-forths in memory (RAM only, forgotten on restart). So she can follow a short conversation, but she doesn't keep a diary.

5.5 How the memory jar works

The memory jar is a **folder of short voice clips in Mat's voice** at `~/axy/memory-jar/`.

- **Tap** the MEMORY JAR button: Axy blushes, wiggles her gills and plays a random clip. It's a shuffle-bag, so there are no repeats until every clip has played once.
- **Rare "jackpot" clips:** start a file name with `rare_` (e.g. `rare_kyoto-promise.m4a`). They play about 1 time in 20.
- **Offline:** the jar needs no internet. When the cloud brain is unreachable and Axy has no saved answer, she says *"I can't reach my cloud pond right now. But I saved something special for you"* and plays a memory.
- **Optional idle surprise:** set `idle_minutes = 90` under `[memory_jar]` and Axy plays a clip on her own after 90 quiet minutes. It's off by default, because surprises at 3 a.m. are less cute.

Recording (do this early - it's the most "you" part): 1. Record 10-20 clips of 5-15 seconds on your phone's voice memo app, in a quiet room. Smile while you talk, and leave half a second of silence at the start and end. 2. Name them `01_goodmorning.m4a`, `02_iloveyou.m4a` ... (.m4a, .mp3, .wav and .ogg all work). 3. Copy them to the Pi from your laptop: `bash scp *.m4a mat@axy.local:~/axy/memory-jar/` Or use a USB stick on the Pi: `lsblk` to find it (e.g. `sda1`), then `sudo mount /dev/sda1 /mnt && cp /mnt/*.m4a ~/axy/memory-jar/ && sudo umount /mnt`. 4. Optional, to make all clips equally loud: in `~/axy/memory-jar/` run `for f in *.m4a; do ffmpeg -i "$f" -af loudnorm -ar 44100 "${f%.*}.wav" && rm "$f"; done` 5. Test: `~/axy/venv/bin/python ~/axy/axy.py --test jar`. The folder is re-read on every press, so no restart is needed.

5.6 Volume and audio devices

- See what Axy found: `~/axy/venv/bin/python ~/axy/axy.py --list-devices`
- Axy picks the **webcam mic** automatically (any card whose name contains C920, C922, Webcam, Logi...), then any other USB mic. For the speaker she prefers a USB speaker, then the Pi's headphone jack.
- To force a device, set `mic_device` / `speaker_device` in `config.toml` to a name from the list, e.g. `"plughw:CARD=C920"`.
- **Hardware volume:** `alsamixer` (press **F6** to choose the speaker's card, arrow keys to set about 60-70%, **Esc** to exit), then `sudo alsactl store` so it's kept after reboot.
- **Software volume:** `tts_volume` (Axy's voice) and `memory_volume` (clips) in `config.toml`. Voice speed: `speaking_speed` (1.1 = a little slower and sleepier).

5.7 Starting, stopping and logs

```
sudo systemctl start axy        # start now
sudo systemctl stop axy         # stop (do this before running --test commands)
sudo systemctl restart axy      # after editing config.toml or axy.env
systemctl status axy            # is she running?
journalctl -u axy -f            # live log: what she heard, what she said, errors (Ctrl+C to
quit)
sudo systemctl disable axy      # stop starting on boot
```

To run by hand for debugging (stop the service first, so they don't fight over the GPIO pins):

```
sudo systemctl stop axy
~/axy/venv/bin/python ~/axy/axy.py -v              # real buttons, verbose log
~/axy/venv/bin/python ~/axy/axy.py --keyboard      # press Enter to talk, or type a sentence
```

5.8 Wi-Fi at her place (no-stress options)

Raspberry Pi OS remembers several Wi-Fi networks and joins whichever is in range. Set up the backups **before** gift day:

1. **Buffalo WMR-300 (best backup).** Add the WMR-300's Wi-Fi to Axy now: `sudo nmcli dev wifi connect "WMR300-SSID" password "WMR300-PASSWORD"` At her place, plug the WMR-300 into her router with one of your LAN cables (router mode; check the WMR-300 manual for its mode switch) and power it by USB. Axy joins it automatically, and you don't need her Wi-Fi password on gift day.
2. **Your phone's hotspot** as another known network (same `nmcli` command). Perfect for the dry run and for emergencies.
3. **A LAN cable** straight from her router to the Pi's Ethernet port. This works instantly, with no setup.
4. Later, add her own Wi-Fi together: SSH in (from your laptop on the same network) and run `sudo nmcli dev wifi connect "HerWiFi" password "..."`.

Hold the MEMORY JAR button for 2 seconds and Axy tells you whether she's online and her IP address.

5.9 Wake word (later, optional)

Push-to-talk is the gift-day default because it's reliable and private: Axy only listens while the button is held. Later you could add a wake word like "Hey Axy" with **openWakeWord** (free, runs on a Pi 4/5). That would mean a small background listener which calls `axy.talk_pressed()` / `talk_released()` around a few seconds of recording. Treat it as a fun "Axy learned a new trick" visit, done at her place, never by taking the gift away.

5.10 Privacy and the third eye (please read)

- Axy **only records while TALK is held** (or when you type on the phone page).
- With **face-follow** on, Axy takes one photo when she talks, to find her face and turn toward her. That photo stays on the Pi (it's overwritten each time).
- A photo is **only sent to the cloud AI when she asks a "looking" question** ("what do you see?", "how do I look?"...). The trigger words are in `look_words`.
- Tell her Axy has a camera "third eye" and how it works. She can switch it off any time: set `[camera] enabled = false`, and a felt "eyelid" over the forehead makes a sweet physical privacy cover.

5.11 How `axy.py` works (for the curious)

`Body` drives servos and LEDs with **gpiozero**. Servos are "detached" after every move, so they don't buzz or jitter while idle. `Audio` records with `arecord` and plays with `aplay` (and `ffmpeg` for .m4a/.mp3). `Ears` is Vosk, `Mouth` is Piper (with espeak-ng as an emergency voice), `ThirdEye` is the webcam (OpenCV, or ffmpeg if OpenCV is missing), `Brain` is Gemini / Groq / Ollama, `MemoryJar` is the shuffle-bag, and a tiny optional web page provides the phone fallback. The **self-tests** (`--test …`) check each part on its own, as listed in Section 7.

6. Fallbacks: when the webcam, mic or internet fails

Axy is built so that **something sweet always happens** when a button is pressed. Here is what she does on her own, and what you can do about it.

6.1 Webcam (camera) fails → Design B, voice-only

Automatic: at start-up Axy checks for the camera (`/dev/video0`). If it's missing she switches to **voice-only mode** (Design B from DESIGNS-AND-FEATURES.md) and says once:

> *"My third eye is a bit sleepy today, but I can still hear you perfectly."*

Chat, gill wiggle, blush, memory jar and the charge tail all keep working. If a photo fails at the moment she asks "what do you see?", Axy says *"I tried to peek with my third eye, but it's all blurry. Ask me again?"*

> **Remember:** the webcam is also Axy's microphone. If the whole webcam dies (not just the picture), she loses her ears too. Then she uses a backup USB mic if one is plugged in (6.2), or the phone page and memory jar. A cheap USB mic in the gift-day kit is good insurance.

What you can do: - Unplug and replug the webcam, then `sudo systemctl restart axy` . Try a different USB port. - Run `--test camera` (Section 7) to see the error. - Make voice-only official: `[camera] enabled = false` . The forehead lens hole becomes a decorative "sleeping third eye". Close it with a felt eyelid. You can upgrade back to Design A later without rebuilding.

6.2 Webcam microphone fails or sounds muffled → backup mic, phone, or memory jar

Axy looks for a microphone in this order: 1. **The webcam's built-in dual mics** (default). 2. **Any other USB microphone** that's plugged in (a cheap USB mic or a USB sound adapter with a mic). Just plug it in and restart Axy: she picks it automatically. You can force it with `mic_device` in `config.toml` . 3. **No microphone at all:** Axy says

> *"Oh no, my little ears aren't working. You can still press my memory button, or talk to me on your phone."*

She then switches on the **phone page** automatically (`[phone_page] mode = "auto"`).

The phone page (talk with your phone): on the same Wi-Fi, open **`http://axy.local:8080`** (or the IP address Axy speaks when you hold the MEMORY JAR button for 2 seconds) on her phone. Type a message, or tap the **microphone key on the phone's keyboard** to dictate. Axy answers **out loud**, with gills and blush, and the reply also appears on the page. There's also a "Memory jar" button. Set `pin = "1234"` (your own number) if you want a PIN. To have the page always on, use `mode = "on"` .

Muffled webcam mic? Usually the wood is covering the mic holes. Enlarge the 3 mm holes in front of the mics (Section 4.4), and speak 30-60 cm from Axy. Check the level with `--test mic`. In `alsamixer` press **F6**, choose the webcam, then **F4** (capture) and raise the level.

6.3 No internet → saved replies, memory jar, optional local brain

Vosk (ears) and Piper (voice) run **on the Pi**, so Axy can still hear and speak offline. Only the "thinking" needs the cloud. When Axy can't reach the internet she does this, in order:

1. **Saved replies:** if what she heard matches a keyword in `[offline_replies]` ("love", "morning", "night", "how are", "thank"...), she answers with that line. *"I love you too. More than all the worms in the whole pond."*
2. **Optional local brain (Ollama)**, if enabled: a tiny model on the Pi itself answers. It's slower and simpler, but genuinely offline.
3. Otherwise: *"I can't reach my cloud pond right now. But I saved something special for you."* Then she plays a **memory-jar clip**.

The **memory-jar button** always works offline.

Optional local brain - only if the hardware allows:

Pi	Local model?
Pi 3 (1GB)	**No.** Too little memory, so use saved replies + memory jar
Pi 4 2GB	Not recommended (very slow, may run out of memory)
Pi 4 4GB / 8GB	Yes: `qwen2.5:0.5b` (faster) or `llama3.2:1b` (a bit smarter, slower)
Pi 5 4GB+	Yes: same models, noticeably quicker

Install with `bash setup.sh --with-ollama` (it checks the RAM and pulls `qwen2.5:0.5b`), or set `[ollama] enabled = true` yourself after installing Ollama. Expect a pause of a few seconds or more per reply, and simpler answers. This is a backup, not the main brain.

Other no-internet fixes: use your phone hotspot, plug in a LAN cable, or bring the WMR-300 (Section 5.8).

6.4 Other things that can go wrong (and what Axy says)

Situation	What Axy does
No API key yet / key placeholder still in `axy.env`	Uses saved replies, otherwise *"Mat hasn't given me my thinking key yet, so I can only blub for now."*
Free-tier limit hit, or the AI service has an error	*"My brain bubbles are a bit busy. Can you ask me again in a minute?"* (If both keys are set, she tries the other provider first.)
She pressed TALK but said nothing / too quiet	*"Sorry, I only heard bubbles. Can you say that again, a little closer?"*
Memory jar folder is empty	*"My memory jar is empty right now. Mat needs to fill it with love."*
Piper voice broken	Falls back to the espeak-ng voice (robotic but working)
GPIO / servo problem	Axy keeps talking without moving (it's logged in `journalctl -u axy`)
Program crash	systemd restarts Axy automatically after 5 seconds

7. Test checklist

Before running any `--test`, stop the service so the two copies don't fight over the pins: `sudo systemctl stop axy`. Then use this shortcut for the commands below:

```
AXY="$HOME/axy/venv/bin/python $HOME/axy/axy.py"
```

7.1 Bench tests - one part at a time (on the desk, before it goes in the body)

✔	Part	Command	Pass when…
☐	Pi boots and is online	`ping -c 3 google.com`	3 replies
☐	Software installed	`$AXY --list-devices`	Lists the webcam as a capture device and `/dev/video0` as a camera
☐	Speaker	`$AXY --test speaker`	A soft two-note "bloop" plays
☐	Voice (Piper)	`$AXY --test tts`	*"Hello [her name]. I am Axy…"* in a clear voice
☐	Webcam mic	`$AXY --test mic`	Your 4-second recording plays back clearly (not muffled)
☐	Speech-to-text	`$AXY --test stt`	Say "what do you see". The printed text is close
☐	Camera	`$AXY --test camera`	Photo saved; then copy it to your laptop with `scp mat@axy.local:/tmp/axy/camtest.jpg .` and check it's sharp and the forehead hole isn't in view. A "face offset" number appears when your face is in view
☐	AI brain	`$AXY --test brain`	"internet: True", your key shows True, and a cute one-line reply
☐	Eye + blush LEDs	`$AXY --test leds`	Both eyes on, off, steady, shimmer; then blush on
☐	Servos	`$AXY --test servos`	Each servo goes left, right, centre, then everything wiggles. **No reboot, no buzzing at rest**
☐	Buttons	`$AXY --test buttons`	"TALK: OK" and "JAR: OK" when you press them
☐	Memory jar	`$AXY --test jar`	Shows the clip count and plays one
☐	Servo power check	Run `--test servos` 3 times in a row	The Pi never reboots, and `--list-devices` still shows the webcam afterwards
☐	Charge tail	Plug her type of phone into the tail	The phone shows charging. Axy doesn't reboot when you plug it in

Run `$AXY --test all` to do everything in one go. It prints PASS/FAIL per part, but still **use your eyes and ears**: a "PASS" for LEDs only means no error happened.

7.2 Full system test (everything assembled, service running)

1. `sudo systemctl start axy` and watch `journalctl -u axy -f` in a second SSH window.
2. ☐ Boot chime plays, then the greeting. (First ever start: the "I just hatched!" line.)
3. ☐ Eyes are slowly "breathing".
4. ☐ **Hold TALK**, say *"Hi Axy, how are you?"*, release. Eyes brighten while you hold, shimmer while she thinks, then she answers in 1-3 sentences, with gills and tail wiggling and cheeks blushing. Typical delay is 2-5 seconds.
5. ☐ Move to one side and talk again: her **head turns toward you**.
6. ☐ Ask *"What do you see?"* while holding a mug: she mentions something real.
7. ☐ Tap MEMORY JAR 5 times: 5 different clips play.
8. ☐ Hold MEMORY JAR for 2 s: she reports "online", her address, camera on, mic on, key ready, number of memories.
9. ☐ Ask 5 questions in a row: it stays snappy, with no "busy" message.
10. ☐ **Unplug the router/Wi-Fi**: ask "Good morning Axy" (saved reply) and something random (cloud-pond line + memory). Plug the network back in. She works again without a restart.
11. ☐ **Unplug the webcam** and restart. She says the sleepy-third-eye line **and** the ears line, because the webcam is also her microphone, and the phone page switches on. Plug in a backup USB mic (if you have one), restart, and check voice-only chat works. Plug the webcam back in and restart.
12. ☐ **Pull the Pi's power** (simulating a blackout), then power on: she restarts by herself within about a minute.
13. ☐ Run for **2 hours** with the hatch closed: touch the body near the Pi. It should be warm, not hot. (Optional check: `vcgencmd measure_temp` stays below about 70°C.)
14. ☐ Phone page: set `mode = "on"`, restart, open `http://axy.local:8080` on a phone, and type a message: she answers out loud. (Set it back to `"auto"` afterwards if you like.)

7.3 Gift-day dry run (a few days before, the full "performance")

Do this exactly as it will happen, ideally on a **different network** (your phone hotspot or the WMR-300), to prove Axy doesn't secretly depend on your home Wi-Fi.

- [ ] Memory jar loaded with the final clips (including at least one `rare_` clip). Each one listened to once on Axy's speaker.
- [ ] `her_name` and `sweet_facts` set. Hatch greeting re-armed: `$AXY --reset-hatch`.
- [ ] Backup Wi-Fi networks saved (WMR-300 + phone hotspot), and both tested.
- [ ] Volume is gentle (comfortable at arm's length, never shouty).
- [ ] Belly note and inside-lid note written. Spare gills in the egg.
- [ ] Egg and nest packed, with Axy in the nest and the cables hidden.
- [ ] **Run the reveal:** power on → wait for the eyes to breathe → open the egg → "I just hatched!" → memory jar tap → push-to-talk "hi" → "what do you see?" → plug a phone into the tail.
- [ ] Time from power-on to the first greeting: _____ seconds (so you know when to plug her in).
- [ ] After the dry run: **`$AXY --reset-hatch`** again, so the real day gets the hatch line.
- [ ] Pack a **gift-day kit**: Pi power supply, servo charger + lead, a LAN cable, the WMR-300 + its USB cable, a spare USB-C cable, a small screwdriver, hot-glue sticks, spare gills, and your laptop or phone with SSH (e.g. Termius) just in case.

8. Troubleshooting

`$AXY` below is the shortcut from Section 7: `AXY="$HOME/axy/venv/bin/python $HOME/axy/axy.py"` (stop the service first with `sudo systemctl stop axy`).

First steps for almost anything: **`journalctl -u axy -n 50`** (the last 50 log lines), then **hold MEMORY JAR for 2 seconds** for her spoken status.

Symptom	Likely cause	Fix
Can't `ssh mat@axy.local`	Pi not on Wi-Fi yet, or `.local` names don't work on your network	Wait 2-3 minutes after the first boot. Find the IP in your router's device list and use it. Re-flash with the Wi-Fi settings re-typed carefully (country **JP**)
`setup.sh` fails at "pip install piper-tts"	32-bit OS	Re-flash **Raspberry Pi OS Lite (64-bit)**. Check with `uname -m`: it must say `aarch64`
`setup.sh` model download fails	Network hiccup	Run `bash setup.sh` again. It continues where it stopped
No sound at all	Wrong output device, or volume at 0	`$AXY --list-devices`. Set `speaker_device` in `config.toml`. `alsamixer` (F6 → your speaker), unmute with **M**, raise the level. Pi 5 has no 3.5 mm jack, so use USB
Sound crackles or cuts out when the gills move	Servos powered from the Pi, or no capacitor	Servos must use the **separate 5V supply**, with grounds joined at pin 9 and a 470-1000 µF capacitor (Section 3.2)
Pi reboots when Axy starts talking	Brownout (servos or a weak Pi supply)	Same as above, plus use the **official** Pi supply. `vcgencmd get_throttled` should print `throttled=0x0`
Servos twitch randomly / don't move	No shared ground, or the wrong GPIO	Join servo GND to Pi pin 9. Check orange wires on pins 11/13/15. Run `$AXY --test servos`
Servo buzzes or strains at the end of travel	Pulse range too wide for your SG90	In `[motion]` change `servo_min_pulse_ms` to 0.6 and `servo_max_pulse_ms` to 2.3, or reduce `gill_swing`
Head turns **away** from her	Servo mounted the other way	`head_invert = true`
Head doesn't turn toward her at all	Face-follow needs OpenCV, light, and a visible face	`sudo apt install python3-opencv opencv-data`. Better front lighting, no bright window behind her. `$AXY --test camera` should print a face offset
Eyes don't light	LED backwards, or a missing resistor/ground	Flip the LED (long leg to the resistor/GPIO side). Check GND to pin 30 and the pins 16/18. Run `$AXY --test leds`
Eyes very dim	White LED on 3.3V	Use pink/orange LEDs, or 100 Ω, or the transistor option (3.4)
TALK button does nothing	Wrong legs on a 4-leg button, or wrong pin	Use **diagonal** legs. GPIO5 = pin 29, GND = pin 25. Run `$AXY --test buttons`
"Sorry, I only heard bubbles" every time	Mic muffled, too quiet, or the wrong mic	`$AXY --test mic`. Enlarge the mic holes in the forehead. Raise the capture level in `alsamixer` (F6 → webcam, F4). Speak 30-60 cm away and keep holding TALK until you've finished
Speech recognised badly	Small English model + room noise	Speak clearly and a little slower. Hold the button 0.5 s before speaking. A larger Vosk model is possible on a Pi 4/5 (e.g. `vosk-model-en-us-0.22-lgraph`)

Symptom	Likely cause	Fix
"Mat hasn't given me my thinking key yet"	Key missing or still the placeholder	`nano ~/axy/axy.env`, paste the key with no quotes/spaces, then `sudo systemctl restart axy`. Check with `$AXY --test brain`
"My brain bubbles are a bit busy"	Free-tier rate limit, a renamed model, or a bad key	The log shows the HTTP error. **429** = wait a minute (or add the 2nd provider's key). **400/404** = check the model name in `[brain]` against the provider's model list. **401/403** = make a new key
"I can't reach my cloud pond"	No internet	Check Wi-Fi (`nmcli dev status`). Use the WMR-300, a phone hotspot or a LAN cable (5.8)
Replies are long or have odd symbols	The model ignored the prompt	Axy already strips emoji/markdown. Make the prompt firmer ("Reply in at most 2 short sentences"). Try `gemini-3.8-flash` for chat
"My third eye is a bit sleepy" at boot	Webcam not detected	Replug, try another USB port, `ls /dev/video*`. If it's dead, set `[camera] enabled = false` (Design B)
Photo is dark/blurry	Low light, or a dirty or blocked lens	Better light. Wipe the lens. Make sure the forehead hole is centred and wide enough
Memory jar plays nothing	Unsupported file, or the wrong folder	Files must be in `~/axy/memory-jar/` with .m4a/.mp3/.wav/.ogg. `$AXY --test jar`. `ffmpeg` must be installed (setup.sh does it)
Memory clips are much quieter than Axy's voice	Phone recordings are quiet	Normalise them (5.5, step 4), or raise `memory_volume` (e.g. 1.6)
Axy reboots when a phone is plugged into the tail	The Pi is powered from the same bank	Power the Pi from its own official supply. The bank is only for the tail (4.8)
Power bank switches itself off	Low-load auto-off	Normal for the tail-only setup: it wakes when a phone is plugged in (press its button if needed). Don't use it for the servos
Body gets hot	Not enough airflow	More vent holes low and high. Add a heatsink (or the Pi 5 active cooler). Keep the bank away from the Pi. Check `vcgencmd measure_temp`
`--test` says "GPIO not available" / "pin in use"	The service is running and holds the pins	`sudo systemctl stop axy` first. Also check you're in the `gpio` group (log out/in after setup)
Phone page won't open	Not on the same network, or the page is off	Both on the same Wi-Fi. Use the IP from the 2-second status. Set `[phone_page] mode = "on"`
After an SD-card scare, odd errors	SD corruption after brownouts	Re-flash and re-run `setup.sh` (keep a copy of `config.toml`, `axy.env` and the clips on your laptop)

Things to back up on your laptop

`~/axy/config.toml`, `~/axy/axy.env` (keep it private), and `~/axy/memory-jar/`. With these three, rebuilding Axy from a fresh SD card takes about 30 minutes.